

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets
Assessment Tool Weightage: 35%

QUESTIONS: 05

Total Marks:50

Annexure A

Pseudocode Writing Task (Algorithm Design)

Criteria	Problem Understanding (2)	Algorithm Design (3)	Use of Control Structures (2)	Pseudocode Structure (1)	Completeness (2)	Total(10)
<i>Weightage</i>	20%	30%	20%	10%	20%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I kamboji Thanuja/19373012 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

K Thanuja

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Annexure A

Q1. Dataset: Online Shopping Transactions

Order ID	Products Purchased
O1	Mobile, Earphones
O2	Mobile, Charger
O3	Earphones, Charger
O4	Mobile, Earphones, Charger
O5	Mobile, Earphones

Question:

Design pseudocode for the **Apriori algorithm** to discover frequent product combinations from the online shopping dataset using a specified minimum support threshold. Clearly illustrate the candidate itemset generation and pruning process at each iteration. (BL3 – 1.05 Marks)

Q2. Dataset: Library Book Borrowing Records

Borrow ID	Books Borrowed
B1	Data Science, Python
B2	Python, Machine Learning
B3	Data Science, Machine Learning
B4	Data Science, Python, Machine Learning
B5	Python, Machine Learning

Question:

Develop pseudocode for the **FP-Growth algorithm** to identify frequent borrowing patterns without generating candidate itemsets. Explain the steps involved in constructing the FP-Tree using the given borrowing records. (BL4 – 1.05 Marks)

Q3. Dataset: Frequently Purchased Services

Service Combination	Support Count
{Internet}	4
{Streaming}	4
{Cloud Storage}	3
{Internet, Streaming}	3

Service Combination Support Count
{Streaming, Cloud Storage} 2

Question:

Write pseudocode to generate **association rules** from the given frequent service combinations. Include the procedures for confidence calculation, rule generation, and filtering based on a minimum confidence threshold. *(BL3 – 1.05 Marks)*

Q4. Dataset: Student Course Enrollment

Student ID Courses Enrolled

S1	Python, Database
S2	Python, AI
S3	Database, AI
S4	Python, Database, AI
S5	Python

Question:

Design pseudocode for **constraint-based frequent itemset mining** where only itemsets containing the course "**Python**" are considered during the mining process. Explain how the constraint affects candidate generation and pruning. *(BL4 – 1.05 Marks)*

Q5. Dataset: Hospital Treatment Hierarchy

Patient ID Treatments Received

P1	Healthcare, Cardiology, ECG
P2	Healthcare, Orthopedics
P3	Healthcare, Cardiology
P4	Healthcare, Cardiology, ECG
P5	Healthcare, Orthopedics

Question:

Develop pseudocode for **multilevel association rule mining** using the treatment hierarchy provided in the dataset. Explain how rules are generated at different hierarchy levels and how concept hierarchies improve pattern discovery.

1. APriori Algorithm - online shopping Transaction

Given dataset

Order	Product
1	Mobile earphones
2	Mobile, charger
3	earphones, charger
4	Mobile, earphones
5	Mobile earphones

Assume minimum support count = 3

Pseudocode

APRIORI (Transaction - Dataset, min support)

1. C_1 - generate all 1-itemsets
2. Count support of every item in C_1
3. $L_1 \leftarrow$ remove items whose support $<$ min-support
4. $k \leftarrow 2$
5. while $L(k-1)$ is not empty:
 - $C_k \leftarrow$ generate candidates from $(k-1)$
 - for each candidate c in C_k :
If any $(k-1)$ -subset of c is not in $L(k-1)$:
Remove c
 - Count support of remaining candidates
 - $L_k \leftarrow$ candidates having support $\geq$ min-support
 - $k \leftarrow k+1$
6. RETURN all frequency itemsets $L_1, L_2, \dots$

2. FP- Growth Library BOOK Borrowing

Dataset

Borrow ID

BOOK

B ₁	Data science, Python
B ₂	Python, machine learning
B ₃	Data science, machine learning
B ₄	Data science, Python, machine learning
B ₅	Python, machine learning

minimum support count = 2
Pseudocode

FP- GROWTH (D, min-support)

1. Scan D and calculate support of each item
2. Remove items whose support < min-support
3. Sort remaining items according to decreasing support
4. Create an empty F-P Tree
5. FOR each transaction T in D:
Remove infrequent items
Sort-items T by frequency
Update node count
6. Create header table linking identical item
7. Mine the FP-Tree recursively:
Find conditional pattern bases
Construct conditional FP-Trees
Generate frequency itemsets
8. RETURN all frequent itemsets.

Constraint - Based frequent Itemset Mining

Dataset

student

S₁

S₂

S₃

S₄

S₅

courses

Python, Database

Python, AI

Database, AI

Python, Database, AI

Python

Final frequent Itemsets

{Python}

{Python, Database}

{Python, AI}

Pseudo code

1. Define constraint:
every itemset must contain "Python"
2. Generate C_1
3. Remove candidate that do not contain Python
4. count support
5. $L_1 \leftarrow$ candidates satisfying min-support
6. FOR $k = 2$ while $L_{(k-1)}$ is not empty:
Generate candidates itemset from $L_{(k-1)}$
For each candidate c :
IF "Python" is not in c
Prune c
ELSE
Add c to L_k
7. RETURN all L_k

3. Association Rule Generation

Dataset

Itemset

support count

{Internet}	4
{streaming}	4
{cloud storage}	3
{Internet, streaming}	3
{streaming, cloud storage}	2

Assume minimum confidence = 60%.

$$\text{Confidence } (A \rightarrow B) = \frac{\text{support}(A \cup B)}{\text{support}(A)} \times 100$$

Pseudo code

GENERATE-RULES (frequent-items, min-confidence)

1. FOR each frequent itemset F:
2. Generate all non-empty subsets A of F
3. $B \leftarrow F - A$
4. $\text{confidence} \leftarrow \text{support}(F) / \text{support}(A)$
5. IF confidence $\geq$ min-confidence:
6. Generate rule $A \rightarrow B$
7. Accept the rule
8. ELSE:
9. Reject the rule
10. RETURN all accepted rule

Multilevel

Association

Rule

mining

Dataset

Patient

Treatments

P₁

Healthcare, cardiology ECG

P₂

Healthcare, orthopedics

P₃

Healthcare, cardiology

P₄

Healthcare, cardiology ECG

P₅

Healthcare, orthopedics

Pseudocode

MULTILEVEL - ASSOCIATION - MINING (0, hierarchy, min-support)

1. Define the treatment hierarchy
2. start from the highest/general level
3. For each hierarchy level:
 - Extract items belonging to that level
 - Count support of each item/itemset
 - Remove items below min-support
 - Generate association rule from frequency items
 - Calculate confidence for each rule
 - keep rules satisfying min-confidence
4. Move to the next, more detailed hierarchy level.
5. Repeat the mining process
6. RETURN frequent patterns & rule from all levels.